

Ali Raza

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SUMMARY

AI/ML Engineer with **3+ years** building and deploying production **machine learning** systems. Core expertise in supervised and unsupervised learning, **NLP**, **computer vision**, and end-to-end ML pipeline development. Experienced integrating ML models into production applications via **FastAPI** with **MLOps** infrastructure including **Docker**, **AWS**, experiment tracking, and model monitoring.

TECHNICAL SKILLS

ML / DL & CV: PyTorch, TensorFlow, Keras, Scikit-learn, HuggingFace Transformers, **Optimization**(LoRA, QLoRA, PEFT) , **ML Models**(XGBoost, LightGBM), fine-tuning, transfer learning, **Computer Vision**(YOLOv8, TransU-Net, U-Net, ViT, SAM, ResNet, DenseNet, BioViL-T, GradCAM, CBAM, OpenCV), **Deep Generative Models**(GANs, VAEs, Probabilistic Models, Diffusion, Mamba)

NLP, LLMs & Data: BERT, LlamaIndex, LangGraph, RAG pipelines, LangChain, FAISS, Pinecone, embeddings, prompt engineering, OpenAI API, Gemini API, Anthropic API, Pandas, NumPy, SQL, PostgreSQL, MongoDB

MLOps, Deployment & Programming: MLflow, Docker, Kubernetes, KubeFlow, Prometheus, Grafana, Evidently AI, FastAPI, CI/CD, GitHub Actions, AWS EC2, inference optimization, GPU deployment, Python, C++, JavaScript, Git

EXPERIENCE

ML Engineer — Freelance | International Clients

2023 – Present

Remote, Faisalabad, Pakistan

- Designed **end-to-end ML pipelines** — preprocessing, training, evaluation, and **scalable production deployment** via FastAPI + Docker with real-time **model performance monitoring**.
- Built and deployed **YOLOv8 computer vision system** — inference-optimized REST API, containerized with Docker, GPU deployment on Vast.ai, integrated with React frontend on Vercel.
- Fine-tuned **LLMs (LoRA/QLoRA)** via HuggingFace PEFT; built **RAG pipelines** with LangChain, FAISS, and Pinecone for document intelligence systems.
- Maintained **MLflow** experiment tracking and model versioning; automated deployments via **CI/CD** pipelines using GitHub Actions.

Data Analyst Intern | TAKMIL

2024

Louisville, Kentucky, USA · Remote

- Cleaned and analyzed large-scale datasets via Python (Pandas, NumPy, SQL); built **Power BI dashboards** — proposed optimizations improving workflow efficiency by **30%**.

RESEARCH & KEY PROJECTS

RadGuard: Multimodal ML System for Medical Imaging — Final Year Thesis

2025–2026

BioViL-T · CXR-BERT · PyTorch · MLflow · Docker · Kubernetes · KubeFlow · FastAPI · AWS EC2

- Built multimodal deep learning system detecting 4 error types in AI-generated chest X-ray reports across 14 conditions on 74,060 MIMIC-CXR samples (**Val F1=0.66**); discovered and fixed 10 critical labeling bugs producing 147K false labels.
- Deployed on **AWS EC2** with Docker + Kubernetes autoscaling, **KubeFlow** retraining pipeline, Prometheus+Grafana monitoring, A/B shadow testing, and **FAISS RAG** chatbot for clinical querying.

RetailSense AI: Production ML Forecasting Platform

2025

LightGBM · XGBoost · ARIMA · K-Means · MLflow · FastAPI · Docker · Kubernetes · AWS EC2 · React

- Built retail demand forecasting system for 54-store grocery chain — trained and compared LightGBM, XGBoost, and ARIMA across 33 product categories; LightGBM achieved **RMSE 336**, **MAPE 8.78%** beating baseline by 55%.
- Tracked **21 MLflow runs** with **statistical A/B testing** via paired t-test ($p=0.0036$); deployed on **AWS EC2** with Docker and Kubernetes; built SHAP explainability and **KS-test drift monitoring** via FastAPI and React dashboard.

Segmentation & Classification of CIEDs — Accepted ACR 2026

2024–2025

TransU-Net · SAM · DINO · ViT · nnU-Net · U-Net

- Designed **TransU-Net** combining transformer attention and U-Net skip connections for automated pacemaker/ICD detection and classification in chest X-rays — fully automatic localization-to-classification pipeline.

ProgressGAN-CXR — Generative AI Research

2025–2026

PyTorch · DCGAN · WGAN-GP · Knowledge Distillation · FastAPI · React · HuggingFace · Docker

- Trained 4 severity-conditioned GAN architectures for continuous COVID-19 disease progression simulation on 21,165 chest X-rays — proposed novel 3-metric evaluation framework (FID, Spearman r, SSIM) showing FID alone is insufficient for medical GANs.
- Deployed full-stack production application — FastAPI inference server on HuggingFace Spaces with Prometheus monitoring, React + TypeScript frontend on Vercel, model weights on HF Hub; live demo at progressgan-cxr.vercel.app.

EDUCATION

BSc Computer Science | **FAST-NUCES**, Faisalabad

2022 – 2026

Co-Head Hackathon ACM Daira 2025 | TAKMIL Data Science Internship 2024

PUBLICATIONS

Ali Raza et al. “Segmentation & Classification of CIEDs in Chest X-rays” — **ACR-2026**